

Protecting Against Ransomware

How WannaCry Spread And What You Can Do About It Today

A hands-on lab walkthrough with screenshots. No malware needed.

The Hospital That Ran Out of Options

May 12, 2017. A hospital in the UK could not open patient records. Monitors that should have shown vitals were showing ransom notes. Surgeries were cancelled. Ambulances were rerouted to other facilities.

Nobody clicked a bad link. Nobody opened a suspicious attachment. A single unpatched Windows machine was sitting on the network with SMB port 445 exposed, and WannaCry found it, exploited it, and used it as a launchpad to infect every other machine it could reach.

Within 24 hours it had spread to 200,000 systems across 150 countries, taking down NHS hospitals, Telefónica, FedEx, and hundreds of smaller organizations.

Here is the part that makes it sting: the patch had been out for two months. Microsoft released MS17-010 on March 14, 2017. Most of the victims just had not applied it.

This post shows you two fixes you can run yourself in a lab environment today, with the exact steps and screenshots to prove they work.

Setting Up Your Lab

Do all of this inside a VM. Take a snapshot before touching anything. Never expose a test machine to the internet.

1. Download and install VirtualBox or Hyper-V (both free)
2. Create a clean Windows 10 or Windows 11 VM from an official ISO
3. Take a snapshot the moment it is ready, before any changes
4. Set the network adapter to Host-Only or Internal Network so the VM stays isolated

You do not need any malware for this. Every test in this walkthrough uses built-in Windows tools and safe actions that anyone can reproduce.

Fix 1: Patch Windows and Remove the SMBv1 Risk

Why This Matters

SMBv1 is a file-sharing protocol that dates back to 1983. It has no encryption, no modern mitigations, and Microsoft has publicly stated it is not safe to use. WannaCry exploited it using EternalBlue, an NSA-developed tool that was leaked by a group called Shadow Brokers. The exploit lets an attacker send specially crafted packets to port 445 and run code remotely, with zero interaction from any user.

Disabling SMBv1 and keeping Windows patched closes that door completely.

Step 1: Check Your Windows Updates

Open Settings, go to Update and Security, then click Windows Update. Hit Check for updates and let everything install. Reboot if it asks you to.

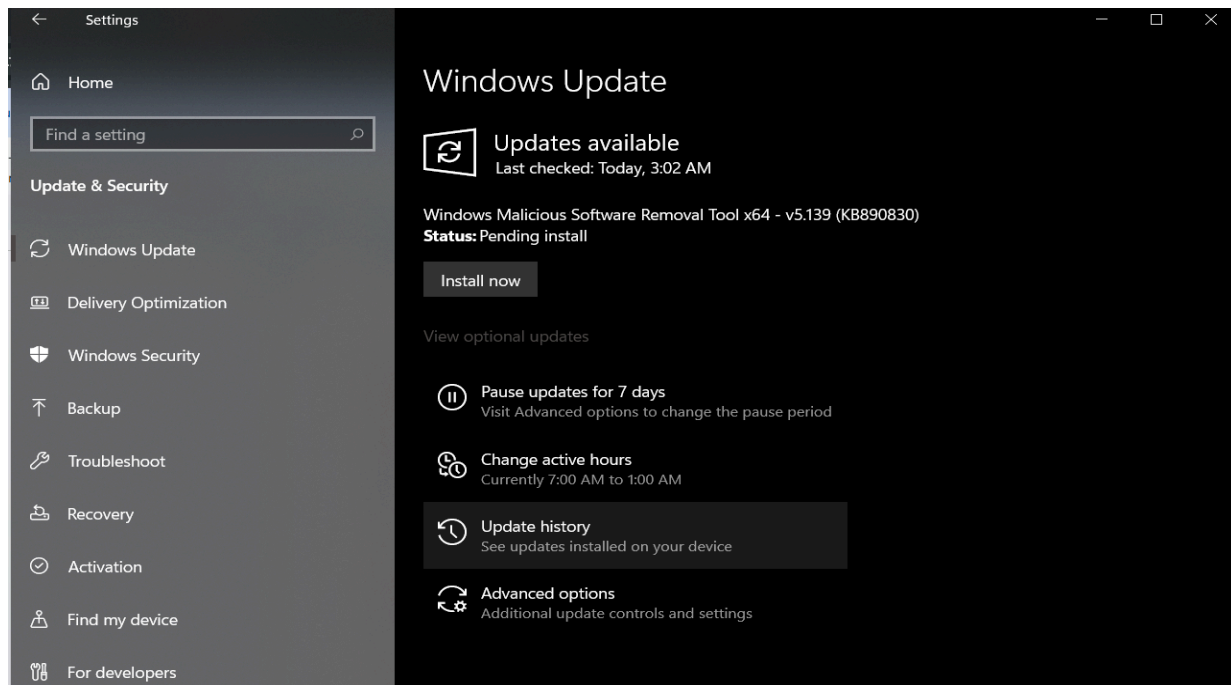


Figure 1: Windows Update page showing pending updates. Check for updates and confirm your last install date.

Step 2: Check if SMBv1 Is Still Running

Open PowerShell as Administrator and run this:

```
PS C:\WINDOWS\system32> Get-WindowsOptionalFeature -Online -FeatureName SMB1Protocol

FeatureName      : SMB1Protocol
DisplayName       : SMB 1.0/CIFS File Sharing Support
Description      : Support for the SMB 1.0/CIFS file sharing protocol, and the Computer Browser protocol.
RestartRequired  : Possible
State            : Disabled
CustomProperties :
                  ServerComponent\Description : Support for the SMB 1.0/CIFS file sharing protocol, and the Computer
                  Browser protocol.
                  ServerComponent\DisplayName : SMB 1.0/CIFS File Sharing Support
                  ServerComponent\Id          : 487
                  ServerComponent\Type       : Feature
                  ServerComponent\UniqueName  : FS-SMB1
                  ServerComponent\Deploys\Update\Name : SMB1Protocol
```

Figure 2: Run as Administrator. Look at the State field in the output.

If State says Disabled you are already protected. If it says Enabled, move to the next step immediately.

Step 3: Disable SMBv1

The GUI route: Control Panel, then Programs, then Turn Windows features on or off. Find SMB 1.0/CIFS File Sharing Support, uncheck it, click OK, and restart.

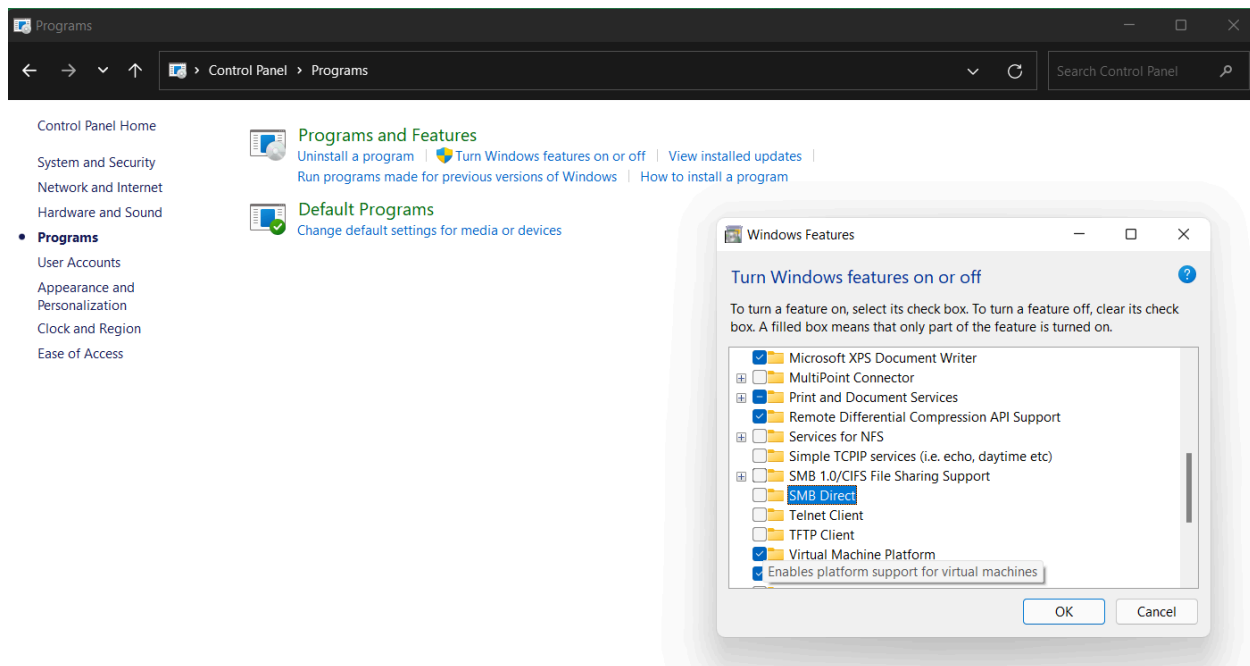


Figure 3: Disabling SMBv1 via Windows Features. Uncheck SMB 1.0/CIFS File Sharing Support and restart.

Or run this in the same PowerShell window:

```
PS C:\WINDOWS\system32> Disable-WindowsOptionalFeature -Online -FeatureName SMB1Protocol -NoRestart

Path      :
Online    : True
RestartNeeded : False

PS C:\WINDOWS\system32> Get-WindowsOptionalFeature -Online -FeatureName SMB1Protocol

FeatureName      : SMB1Protocol
DisplayName      : SMB 1.0/CIFS File Sharing Support
Description      : Support for the SMB 1.0/CIFS file sharing protocol, and the Computer Browser protocol.
RestartRequired  : Possible
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CustomProperties :
    ServerComponent\Description : Support for the SMB 1.0/CIFS file sharing protocol, and the Computer
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    ServerComponent\DisplayName : SMB 1.0/CIFS File Sharing Support
    ServerComponent\Id         : 487
    ServerComponent\Type       : Feature
    ServerComponent\UniqueName  : FS-SMB1
    ServerComponent\Deploys\Update\Name : SMB1Protocol
```

Figure 4: SMBv1 confirmed Disabled after running the PowerShell disable command

Fix 2: Turn On Controlled Folder Access

What This Actually Does

Patching removes the entry point. Controlled Folder Access is what stops ransomware from doing damage even if it somehow gets onto the machine anyway.

It works by putting a guard on specific folders. Only apps that you explicitly approve can write to those locations. Anything else, including ransomware that has somehow slipped past your defenses, gets blocked and logged.

Step 1: Find the Setting

Open Start, then Windows Security, then Virus and threat protection. Scroll down until you see Ransomware protection and click Manage ransomware protection.

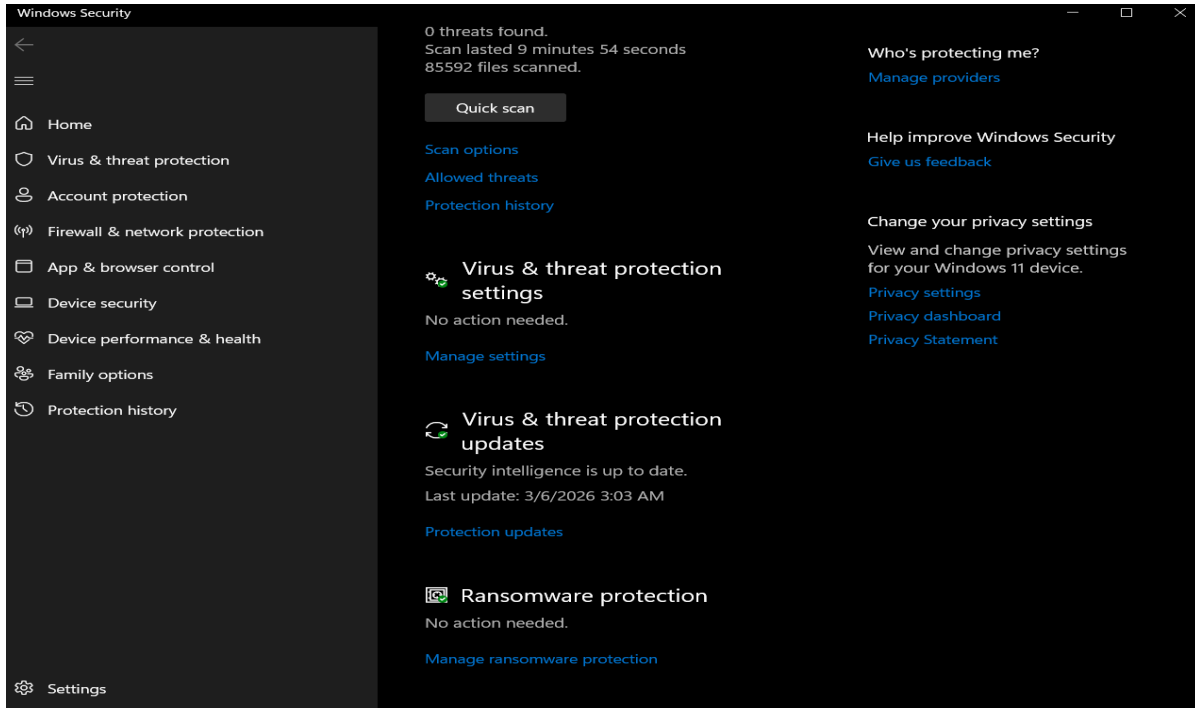


Figure 5: Windows Security main dashboard. Navigate to Virus and threat protection to find Ransomware protection.

Step 2: Enable the Toggle

Flip Controlled folder access to On. It is off by default on almost every Windows install, which means most machines have no protection at this layer.

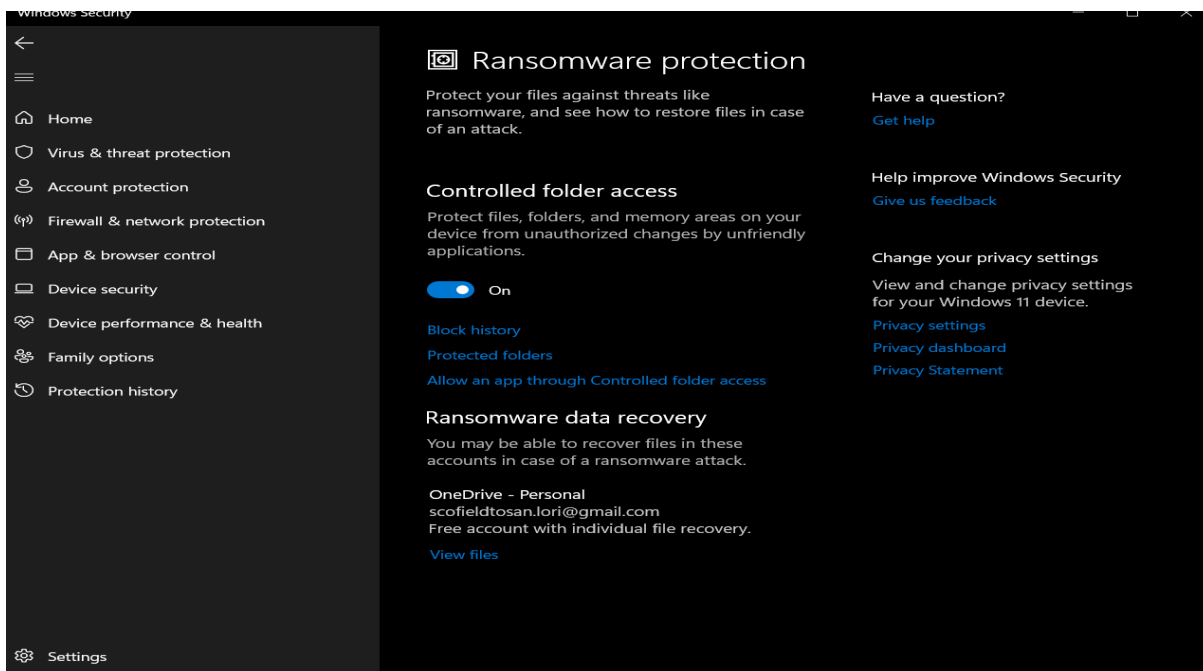


Figure 6: Controlled Folder Access toggle switched On. Off by default on most Windows installs.

Step 3: Add a Protected Folder

Click Protected folders, then Add a protected folder, and create a test path. In your VM, first run this to create the folder:

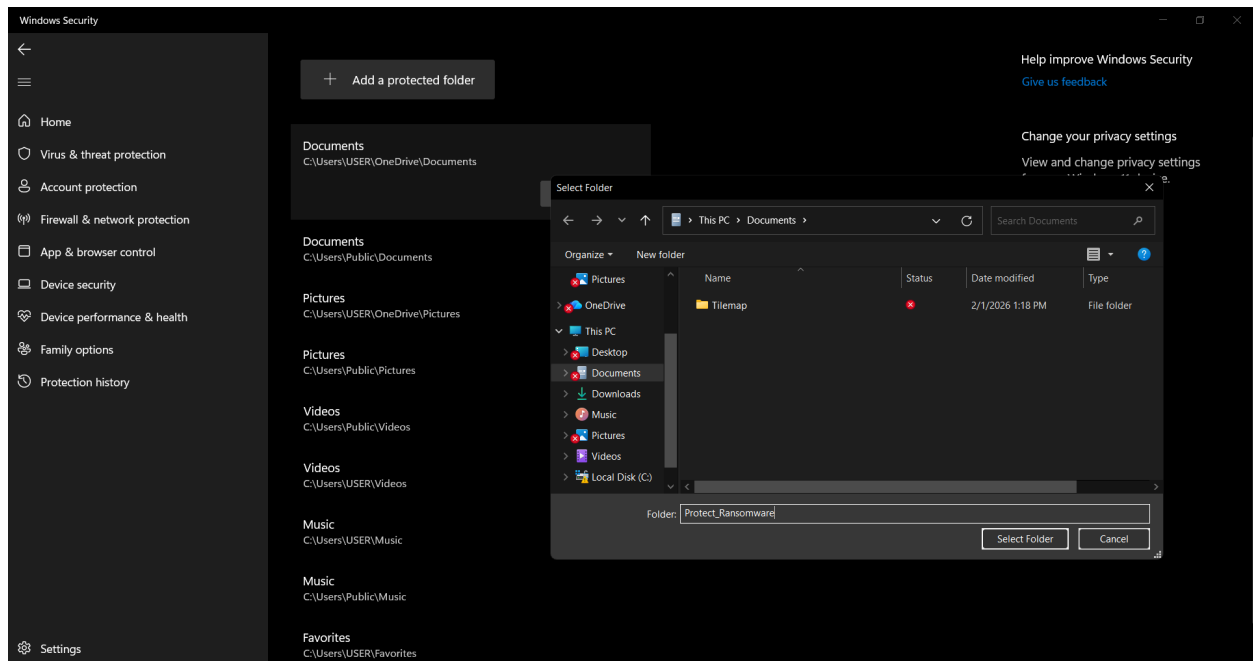


Figure 7: Adding Protect_Ransomware as a protected folder through the Windows Security UI.

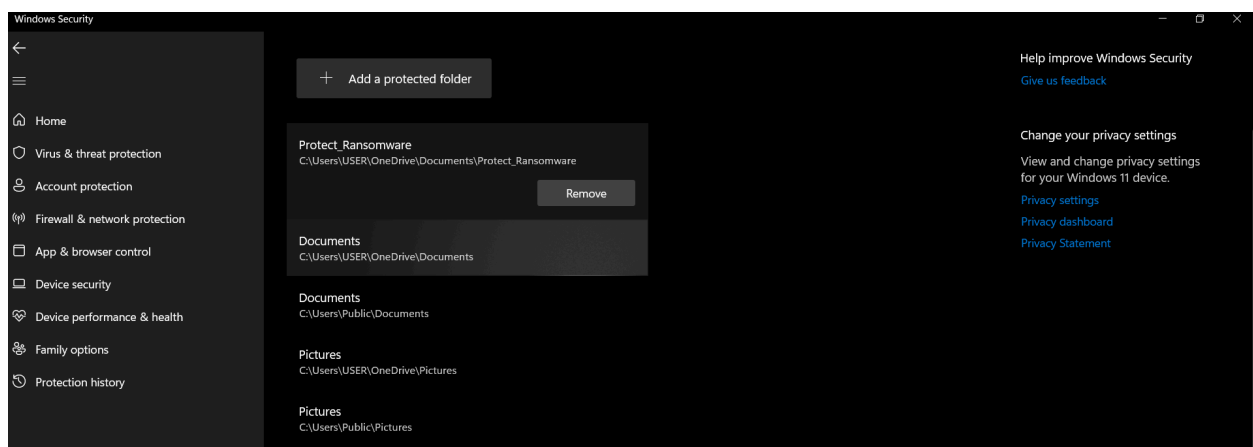


Figure 8: Protected folder successfully added and visible in the protected folders list.

Step 4: Prove It Works

Open PowerShell as a regular user, not as Administrator. That matters because running as admin can bypass certain protections, and the whole point is to simulate what an untrusted process looks like to Windows.

Run this in your PowerShell window:

```
PS C:\Users\USER> # This will try to write a file into your protected folder as an untrusted process
>> $path = "C:\Users\USER\OneDrive\Documents\Protect_Ransomware\stuff.txt"
>> [System.IO.File]::WriteAllText($path, "test")
Exception calling "WriteAllText" with "2" argument(s): "Access to the path is denied."
At line:3 char:1
+ [System.IO.File]::WriteAllText($path, "test")
+ ~~~~~
+ CategoryInfo          : NotSpecified: (:) [], MethodInvocationException
+ FullyQualifiedErrorId : UnauthorizedAccessException
```

Figure 9: PowerShell session showing "Access to the path is denied" — Controlled Folder Access blocking an unauthorised write in real time.

"Access to the path is denied."

That is the block. That is Windows Defender stopping an unauthorised write to your protected folder in real time, exactly what it would do to a ransomware process trying to encrypt your files.

Go to Windows Security, then Virus and threat protection, then Protection history to find the blocked event entry. That entry is your proof of concept.

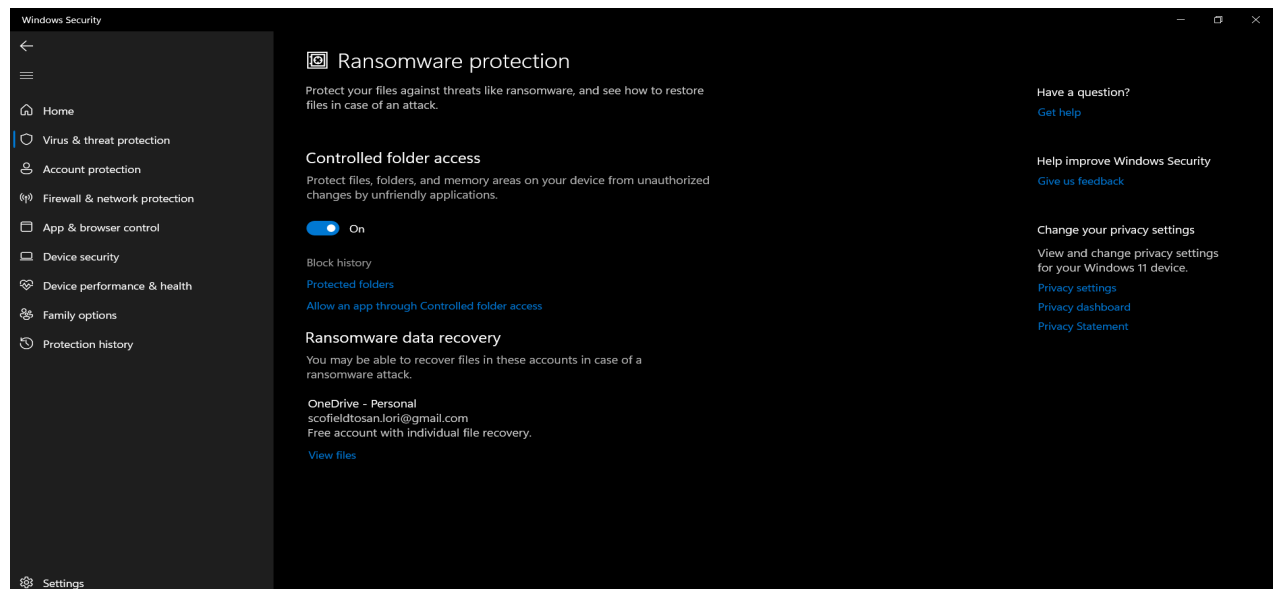


Figure 10: Ransomware protection page confirming Controlled Folder Access is active and On.

What I Found in My Lab

Honestly, this one gave me a bit of trouble before it clicked.

SMBv1 came back Disabled on the fresh Windows 11 install expected, but still good to confirm with your own eyes rather than just assuming. If you are testing on an older build, do not assume. Run the command.

The Controlled Folder Access test is where I spent the most time. I created the Protect_Ransomware folder inside Documents, added it through Windows Security, then opened a regular PowerShell session and tried to write directly into it. First attempt failed because the file did not exist yet. Second attempt hit the wall "Access to the path is denied." Clean block.

One thing worth being honest about: Protection history showed "No recent actions" even after the block registered in PowerShell. The UI did not log it visibly. If you hit the same thing, do not panic, the terminal error is your proof. The block happened. Windows just did not feel like reporting it to the dashboard that day.

Total time for both fixes including a reboot: about 15 minutes. Less time than most WannaCry victims spent staring at that ransom note.

What to Do After This

Set Windows Update to automatic. Push the SMBv1 disable via Group Policy so no one in your environment quietly re-enables it. Back up to an offline or immutable destination, Controlled Folder Access is a layer of protection, not a backup strategy. Test your restores, because a backup you have never actually restored is just a folder you feel good about. And run Microsoft Attack Surface Analyzer on anything critical to find the gaps you did not know were there.

WannaCry's real lesson was not technical. The patch existed. The problem was that nobody had a process that got patches applied fast enough. Build that process before something forces you to.